OSCILLOSCOPE USING ESP32

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INTRODUCTION

This project presents a compact, low-cost, and portable digital oscilloscope developed using an ESP32 microcontroller. It enables waveform visualization with real-time voltage and frequency measurement on a web-based display. The system offers better flexibility and cost-efficiency compared to traditional CROs and DSOs

METHODOLOGY

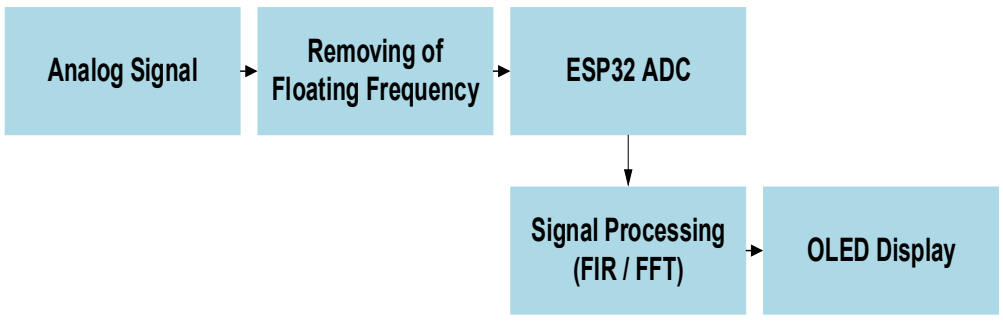

Figure 1: Block Diagram

- Analog Signal – The input waveform (e.g., sine, square, or any analog signal) is applied to the circuit for measurement.
- Removing of Floating Frequency – A simple filter circuit is used to remove unwanted noise or high-frequency fluctuations from the signal, ensuring clean input to the ADC.
- ESP32 ADC – The ESP32's built-in Analog-to-Digital Converter (ADC) converts the filtered analog signal into a digital form for further processing.
- Signal Processing (FIR / FFT) – The ESP32 processes the digitized signal using techniques like Finite Impulse Response (FIR) filtering and Fast Fourier Transform (FFT) for waveform analysis and frequency response.
- OLED Display – The processed signal is displayed on the OLED screen in real-time as a waveform along with voltage and time information, similar to a digital oscilloscope.

WORKING OF MODEL

The input analog signal is first passed through a filter circuit to remove unwanted noise and floating frequencies. The clean signal is then fed to the ESP32's ADC, which converts it into digital form. Inside the ESP32, the signal is processed using algorithms like FIR filtering and FFT to analyze waveform and frequency components. Finally, the processed data is displayed on the OLED screen in real-time, showing parameters like voltage, time scale, and waveform shape similar to a traditional oscilloscope.

PROTOTYPE

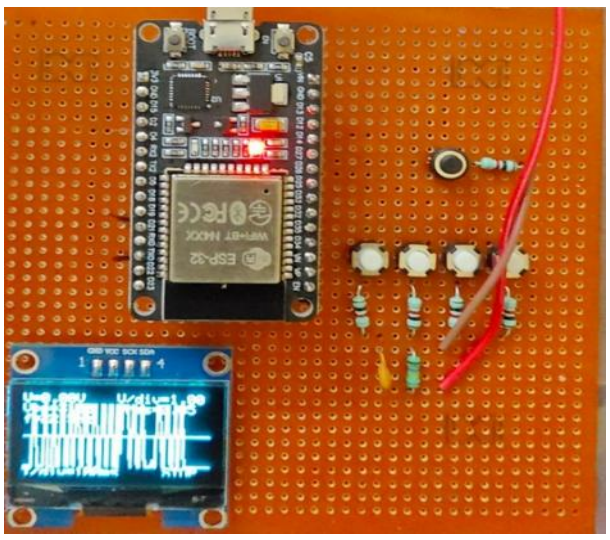

Figure 2: Prototype of model

RESULTS

Waveform	Freq (Hz)	Measured (Hz)	Error (%)
Sine	100	99.2	0.8
Sine	1,000	995	0.5
Square	500	496	0.8
Triangle	200	198	1.0
Sine (Noisy)	2,000	1,988	0.6

Table 1: Results

CONCLUSION

The developed system provides an affordable and portable alternative to traditional CROs and DSOs. It efficiently captures and analyzes real-time signals using features like FIR filtering and FFT for accurate waveform and frequency observation. The system maintains good measurement accuracy and low power consumption, making it ideal for educational and basic testing purposes. Overall, the project demonstrates the potential of microcontroller-based instruments for modern, low-cost electronic measurement applications.

REFERENCES

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